MIL-H-27893D(GL)
4 February 1986
SUPERSEDING
MIL-H-27893C
25 March 1981

MILITARY SPECIFICATION

HARNESSES, PARACHUTIST'S, INDIVIDUAL EQUIPMENT PACK ("H" - TYPE)

This specification is approved for use by Natick Research, Development, and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers two types of general purpose harnesses, having an "H" design and made of nylon webbing. The harnesses are equipped with quick-fit adapters, buckles and snaps to secure it around the load and to the T-10 parachute harness worn by the parachutist.

1.2 Classification. The harnesses shall be of the following types as specified (see 6.2):

Type I - "H" Harness
Type II - H-Harness, Single Point Release

2. APPLICABLE DOCUMENTS

2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

SPECIFICATIONS

FEDERAL

T-T-871	-	Twine, Cotton, Wrapping
PPP-B-26	-	Bag, Plastic, Polyethylene
PPP-B-601	-	Boxes, Wood, Cleated-Plywood
PPP-B-636	-	Boxes, Shipping, Fiberboard

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal, (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1670

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STANDARDS

FEDERAL

FED-STD-751 - Stitches, Seams and Stitching

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads

DRAWINGS

U.S. ARMY NATICK RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER

11-1-147 - Harness Parachutist's, Individual Equipment Pack; ("H" - Type), Assembly and Details, Type I
11-1-3008 - Harness, Parachutist's Individual Equipment Pack; Single Point Release, Type II, Assembly
11-1-3009 - Harness
11-1-3010 - Strap, Leg, Release Assembly
11-1-3011 - Handle Release Assembly
11-1-3012 - Strap, Harness, Attaching

(Copies of documents required by the contractor in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association documents are available for reference from libraries. They are also distributed among using technical groups and Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

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3. REQUIREMENTS

3.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.3, 6.2, and 6.3).

3.2 Materials. Materials (see 6.6), shall conform to the drawings listed in section 2, to the subsidiary specifications, standards and drawings applicable thereto, and as specified herein.

3.3 Construction and assembly. The construction shall conform to the requirements of this document and to the designs, details, and dimensions shown or specified on the applicable drawings.

3.3.1 Stitching. All stitching shall conform to FED-STD-751.

3.3.1.1 Type 301 and 308 stitching. Ends of all type 301 stitching shall be backstitched or overstitched not less than 1/2 inch past ends of the defect in accordance with the required stitch pattern. Ends of type 308 stitching shall be secured by crowding a minimum of three stitches (accomplished by not allowing the materials to move) at the beginning and end of the stitching Thread tension shall be maintained so that there will be no loose stitching resulting in the puckering of the material sewn. The lock shall be embeded in the materials sewn.

3.3.1.1.1 Repairs of type 301 stitching. Repairs of type 301 stitching shall be as follows:

a. When thread breaks or bobbin run-outs occur during the stitching, the stitching shall be repaired by restarting the stitching a minimum of 1 inch (1/2 inch box-x and W-W stitching) back of the end of the stitching. 1/

b. Thread breaks or two or more consecutive skipped or run-off stitches noted during inspection of the item (in-process or end item) shall be repaired by overstitching. The stitching shall start a minimum of 1 inch in back of the defective area, (1/2 inch for box-x and W-W stitching), continue over the defective area and continue a minimum of 1 inch (1/2 inch for box-x and W-W stitching) beyond the defective area onto the existing stitching. Loose or excessively tight stitching shall be repaired by removing the defective stitching, without damaging the materials, and restitching in the required manner. 1/

1/ When making the above repairs, the ends of the stitching are not required to be backstitched.

3.3.1.1.2 Repair of 308 stitching. Repairs of type 308 stitching shall be in accordance with 3.3.1.1.1a and 3.3.1.1.1b, except that the stitching shall overstitch the defective area a minimum of three stitches onto the existing stitches.

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3.3.1.1.3 Automatic stitching. Automatic sewing machines may be used to perform any of the stitch patterns provided the requirements for the stitch pattern, stitches per inch, size and type of thread are met; and at least three tying, overlapping or backstitches are used to secure the ends of the stitching.

3.4 Thread ends. All thread ends shall be trimmed to 1/4 inch maximum length.

3.5 Setting of grommets. Holes shall be prepunched to receive the grommets. Holes prepunched to receive the grommets shall be smaller than the outside diameter of the grommet barrel so that the barrel must be forced through the hole. The grommet shall be securely clinched without cutting the adjacent material.

3.6 Finishing of jacketed wire rope ends. The ends of the nylon jacketed wire rope shall be coated with a melted nylon II or nylon 6/6 powder.

3.6.1 Apparatus for sealing. The following apparatus' are required for sealing nylon jacketed wire rope ends.

a. Induction coil (2-4-inch) with automatic timer control and power source sufficient to melt the nylon jacketed wire rope nylon material.

b. Electric melting pot with automatic temperature control for melting of specified nylon coating material.

3.6.2 Heat sealing conditions. The following heat sealing conditions are required to provide an acceptable sealed and formed jacketed wire rope coated end (see 6.5).

	Minimum	Maximum
	390°	420°
Heating of jacketed wire rope end.	10 seconds	10 seconds

3.7 Repairs. Repairs such as mends, darns, patches or splices are not permitted on the harness.

3.8 Replacement of defective components. During the spreading, cutting and manufacturing process, components having material defects or damages that are classified as defects in 4.4.3, shall be removed from production and replaced with nondefective and properly matched components.

3.9 Workmanship. The finished and assembled harness shall conform to the quality of product established by this document. The occurrence of defects shall not exceed the applicable acceptable quality level.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the document shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 6.2), it shall be examined for the defects specified in 4.4.3 and 4.4.4. The presence of any defect shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.4.2 In-process inspection. Inspection shall be made at any point or during any phase of the manufacturing process to determine whether cut lengths, cut parts, markings for location of components, and location of assembled component parts are in accordance with specified requirements. Whenever nonconformance is noted, corrections shall be made to the parts affected and lot in process. Parts which cannot be corrected shall be removed from production.

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4.4.3 End item visual examination. The end item shall be examined for the defects listed in table I. The lot size shall be expressed in units of harnesses. The sample unit shall be one harness. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 1.5 for major defects and 6.5 for total (major and minor combined) defects.

TABLE I. End item visual defects

Examine	Defect	Classification Major	Minor
Webbing or tape	Not specified type or class.	X	
	Color not as specified		X
	Any hole, cut, or tear.	X	
	Edge frayed, beaded, corded, or torn.	X	
	Ends not seared as specified.		X
Fastener tape	Cut or tear.	X	
	Hooks crushed affecting closure.	X	
	Any fastener tape reversed, i.e., pile tape attached where hook tape is required or visa versa.		X
	Not specified type and color.	X	
Cord	Not specified type and color.	X	
	Ends not seared as specified.		X
Hardware	Fractured, malformed, or bent	X	
	Burrs or sharp edges.	X	
	Protective finish omitted or not as specified, any corroded area.	X	
	Not specified type, size or style.	X	
	Adapter attached with friction grip facing wrong direction	X	
Plastic buckle	Not as specified.	X	
	Broken, cracked, chipped, or malformed, failing to serve intended purpose; burr, sharp edge or flash	X	
	Color and finish not as specified.		X
	Not functioning properly, failing to effect a secure closure or open freely.	X	

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TABLE I. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Jacketed wire rope	Not specified type or size.	X	
	Not as specified.	X	
	Ends not nylon coated and sealed completely exposing wire material.	X	
Swagging wire rope sleeve	Missing.	X	
	Not specified type or size.	X	
	Not swagged as specified.		X
	One sleeve swag omitted.		X
	Two sleeve swags omitted.	X	
	Swagged excessively tight, cutting wire rope material.	X	
	Swagged loosely, permitting sleeve to rotate freely.	X	
Brass grommets	Clinched excessively tight, cutting adjacent material.	X	
	Insecurely clinched to a degree that grommet may be detached from material.	X	
	Washer installed on incorrect side of material.		X
	Barrel split.		X
	Loosely clinched, permitting grommet to rotate freely within the hole.		X
	Washer omitted.	X	
Thread	Not specified type, class, or size.	X	
	Not specified color.		X
Open seams	1/2 inch or less.		X
	More than 1/2 inch.	X	
	NOTE: A seam shall be classified as open when one or more stitches joining a seam are broken or when two or more consecutive skipped or runoff stitches occur.		
Runoffs (see open seams)			
Stitch type	Wrong seam or stitch type	X	

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TABLE I. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Stitches and stitching	Stitching pattern not as specified.	X	
	Any row of stitching missing.	X	
	One stitch per inch under or over minimum and maximum specified.		X
	Two or more stitches per inch under or over minimum and maximum specified.	X	
	Thread breaks, bobbin runouts, or two or more consecutive skipped or run-off stitches not overstitched or restitched as specified.	X	
	Ends of stitching not backstitched or secured as specified.		X
	Tension loose resulting in loose top or bottom thread, or tight tension causing puckering of material at stitch line.	X	
	Needle chews or bunched stitching.	X	
Components and assembly	Any component part omitted or not as specified or required operation omitted (unless otherwise classified herein).	X	
	Any component misplaced or not assembled as specified.	X	
	Any darn, splice, or other unauthorized repair.		X
Identification marking	Omitted, illegible, incorrect, misplaced, or size of characters not as specified.	X	
	Color not as specified.		X
Cleanness	Dirt, grease or oil or foreign matter clearly noticeable.		X
	Thread ends not trimmed.		X
Construction and assembly	Any component missing, or not assembled or joined as specified.	X	

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TABLE I. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Marking	Omitted; incorrect; illegible; wrong location, size or color.		X
Workmanship	Grease, oil, ink smears, dirt or other foreign matter; thread ends not trimmed throughout.		X

4.4.4 End item dimensional examination. The end item shall be examined for conformance to the specified dimensions. Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of harnesses. The sample unit shall be one harness. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

4.4.5 Packaging inspection. An inspection shall be made to determine that the preservation, packing, and marking comply with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully packaged. The lot size shall be the number of containers in the inspection lot. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Any component missing, damaged or not as specified
Workmanship	Inadequate application of component such as incomplete closure of container flaps, loose strapping, or taping, or inadequate stapling. Bulged or distorted container
Contents (unit pack and exterior container)	Number per unit pack or exterior container not as specified.

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4.4.6 Palletization examination. An examination shall be made to determine that the palletization complies with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one palletized unit load fully packaged. The lot size shall be the number of palletized unit loads in the inspection lot. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirement.
Palletization	Not as specified. Pallet pattern not as specified. Interlocking of loads not as specified. Load not bonded with required straps as specified.
Weight	Exceeds maximum load limits.
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A. Each harness shall be compactly folded in such a manner that the quick-fit adapters and snap hooks are cushioned within the folds of the nylon webbing. Care shall be given when folding the type II harness so that the release wires are not bent back on themselves causing a permanent kink. The folded harness shall be securely tied with twine conforming to type I or II, 8-ply of T-T-871 and then inserted in a close-fitting 0.003 inch thick polyethylene bag conforming to type I or II, grade A, style 1 of PPP-B-26. Prior to or during the closure operation, excess air within the bag shall be expelled.

5.1.2 Commercial. Each harness shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B or Commercial as specified (see 6.2).

5.2.1 Level A packing. Forty harnesses, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container, conforming to style RSC-L, grade V2s of PPP-B-636; or overseas type, style A or J, type 2 load of PPP-B-601, as specified (see 6.2). The inside of each fiberboard shipping

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container shall be fitted with a box liner conforming to type CF, class weather-resistant, variety DW, grade V15c of PPP-B-636. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced as specified in the appendix of PPP-B-636, except the inspection shall be in accordance with 4.4.5.

5.2.2 Level B packing. Forty type I harnesses or twenty type II harnesses preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC-L, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. The inside of each shipping container shall be fitted with a box liner conforming to type CF, class domestic, variety DW, grade 275 of PPP-B-636. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636, except the inspection shall be in accordance with 4.4.5.

5.2.2.1 Weather-resistant containers. When specified (see 6.2), the shipping container shall be a grade V3c, V3s or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.5.

5.2.3 Commercial packing. Harnesses, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Palletization. When specified (see 6.2), harnesses packed as specified in 5.2, shall be palletized on a 4-way entry pallet in accordance with load type I or Ia of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L or film bonding means O or P. Pallet patterns shall be in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the pallet patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contracting officer.

5.4 Marking. In addition to any special marking required by the contract or purchase order, shipping containers and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

5.4.1 Polyethylene bagged unit packs. Polyethylene bagged unit packs shall have the required information legibly printed or stamped in black directly on the bag across the center face or on a white paper label inserted within the bag so as to permit ready identification. The bag or label shall indicate the following instructions:

Stock Number
Nomenclature
Quantity

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6. NOTES

6.1 Intended use. The harnesses are intended for use by individual parachutists for securing individual combat equipment during airborne jump operations. The harness with the secured load is attached to the parachutist harness D-rings with the attaching strap snaps.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, name, and date of this document.
b. Type required (see 1.2).
c. When a first article is required (see 3.1, 4.3, and 6.3).
d. Selection of the applicable levels of preservation and packing (see 5.1 and 5.2).
e. Type of shipping container desired for level A packing (see 5.2.1).
f. When weather-resistant grade shipping containers are required for level B packing (see 5.2.2.1).
g. When palletization is required (see 5.3).

6.3 First article. When a first article is required it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample consisting of one complete harness. The contracting officer should include specific instructions in all acquisition documents regarding arrange-ments for inspection and approval of the first article.

6.4 Color samples. For access to standard shade samples of webbing, cord and thread, address the contracting activity issuing the invitation for bids.

6.5 Source for heat sealing. A source for the requirement of sealing and forming the ends of the nylon jacketed wire rope is Boyd Coatings Research Co. Inc., 51 Parmenter Road, Hudson, MA 01749.

6.6 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the document (see 3.2).

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:

Army - GL

Review activity:

Army - AV

Preparing activity:

Army - GL

Project No. 1670-A732

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-H-27893D(GL)	Harnesses, Parachutist's, Individual Equipment Pack ('H' - Type)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NATICK OP, 20 Nov 84

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